

Yaswanth Deevi

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SUMMARY

Computer Science Master's Student at BTH with substantial experience in deep learning, computer vision, and processing multi-modal sensor streams. Skilled in transfer learning, model compression (quantization/pruning), and porting machine learning models on heterogeneous architectures/hardware. Familiar with using camera, pose, and raw signal sensor streams and aggregating them in a unified decision-making pipeline.

EXPERIENCE

Computer Vision Engineer (Pose & Interaction)

June 2026 – Aug 2026

SolarFi's FairWAI, Boston, USA

- Used a perception pipeline, which includes multi-person pose estimation, in two radically different backbones and hardware backends: Vision on iPhone Neural Engine and INT8 quantized MoveNet Lightning on CPU, with exactly the same constraints on accuracy and latency, without starting from scratch.
- Designed a hardware agnostic interaction layer: temporal pose-to-primitives mapping system (geometric heuristics + calibrated MLP), which took the output of keypoints from both backbones and generated a common downstream representation.
- Iteratively created a compact budgeted textual summary that combines multi-person pose with the interaction state in one unified interface to be used by vision-language model – quite a challenge to map different sensor inputs into the shared interface for reasoning.
- Added calibrated confidence information and temporal smoothing so that the signal from either backend did not interfere with the fused representation due to noise.
- Prototyped a YOLOv8 and solvePnP-based 3D head-pose estimation enhancement targeting two underperforming interaction primitives, layered onto the production pipeline (validated at macro F1 0.853 across 148 labeled clips) through an isolated wrapper architecture that kept the existing system unaffected.
- Built a standalone rule-only A/B evaluation harness to test the new signal against ground-truth data at zero retraining cost, and documented the resulting comparison, including a negative result on one primitive, in a fairness-aware technical report.
- Conducted per-primitive precision/recall analysis and skin color/ethnicity disparity analysis on a set of 50 clips for accuracy and fairness validation across demographics.
- Collaborated within a 3-person team by sharing events and track IDs and having a hardware integration meeting every two weeks.

Research Intern

Jul 2025 – Sep 2025

Defence Research and Development Organisation (DRDO), Hyderabad

- Developed, in collaboration with senior researchers, a novel approach towards solving the P-Center Facility Location problem where fitness is estimated through the use of a pre-trained deep neural network surrogate rather than fitness evaluation that is knowledge transfer from an expensive exact model to a cheaper approximation. This was done with the effect of reducing generation-wise cost and speeding up the process of convergence by **an order of magnitude**.
- Developed novel adaptive Gaussian mutations through sigma scaling based on the diversity of the population in order to balance exploration with exploitation without having to re-evaluate the whole search space.
- Performed ablation studies where full-exact, surrogate-only, and hybrid evaluations were compared in order to measure the tradeoff between accuracy and efficiency in replacing an exact computation model with a learned one.
- Collaborated with senior scientists in order to document methods and report results.

EDUCATION

M.S. in Computer Science & Engineering

Expected Jul 2027

Blekinge Tekniska Högskola, Sweden

Relevant Coursework: Research Methodologies, Advanced Machine Learning, Deep Learning, Decision Support Systems

B.Tech in Computer Science & Engineering

Dec 2025

Jawaharlal Nehru Technological University Hyderabad

Relevant Coursework: Machine Learning, Data Structures and Algorithms, Computer Networks, Object Oriented Programming, Python Programming.

TECHNICAL SKILLS

Computer Vision & Perception: CNNs, Pose Estimation (Apple Vision, MoveNet), YOLOv8, Face Detection/Alignment (MTCNN), Frame-level Classification, Multi-modal Signal/Image Processing, OpenCV, ONNX Runtime

Deep Learning Frameworks: PyTorch, TensorFlow, Keras, RNNs, BiLSTM with attention

Deep Learning & Transfer: Transfer Learning, Model Quantization (INT8/4-bit), Model Pruning, Knowledge

Distillation-adjacent surrogate modeling, ONNX, Core ML, Cross-architecture deployment (Neural Engine & CPU)
Optimization & Efficiency: Evolutionary Algorithms, Surrogate Modeling, Combinatorial Optimization, Meta-heuristics – with a focus on reducing computational cost and hardware-aware trade-offs

Programming: Python, C++, SQL, Bash scripting

Machine Learning & Statistics: Scikit-learn, XGBoost, LightGBM, Gradient Boosting, Random Forests, PCA, SHAP, Statistical Hypothesis Testing

Data & Analytics: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Time-Series Analysis

MLOps & Tools: Git, Docker, Linux, AWS, Jupyter, VS Code, Flask Deployment

PROJECTS

Deepfake Detection System *(From coursework)*

GitHub Repository

- Used transfer learning approach for fine-tuning the pre-trained EfficientNet-B0 model on the task of binary classification of deepfakes based on Celeb-DF dataset.
- Developed a video inference pipeline on per frame basis using MTCNN along with dynamic frame skipping to balance accuracy with efficiency.
- Deployed an end-to-end Flask web application and analyzed the quality of the solution in terms of metrics such as F1 score, ROC AUC, confusion matrix etc. with respect to different compression rates and lighting conditions.

Neural Digital Pre-Distortion (OpenDPD)

GitHub Repository

- Developed two new backbones (MLP with skip connections and BiLSTM with scaled dot product attention) to perform the same signal processing task using paired IQ sensor data to check their applicability in terms of hardware constraints that act as the first-hand experience of working with different architectures on the same problem.
- Developed an indirect learning training pipeline using sliding window generation of samples for multi-modal signal data, which can be used for any sequence of sensor data.
- Evaluated performance by using EVM (%) and NMSE (dB) for both the backbones, and stated the reason why BiLSTM outperformed MLP due to high effective memory that acts as a comparison technique between different architectures.

Hybrid Optimization for P-Center Problems *(From research internship)*

GitHub Repository

- Created an evolutionary search algorithm based on surrogates that uses feed-forward DNNs as a surrogate for the fitness function to eliminate exact distance computations.
- Achieved 8–10 times faster convergence compared to standard genetic algorithms and obtained solutions that differed from the optimal by only 1.5% on standard OR-Lib and TSPLIB problems.
- Performed ablation studies using exact computation, surrogate, and surrogate+exact computation approaches to assess the balance between accuracy and efficiency.

Anomaly Detection in Building Heating Systems *(From coursework)*

GitHub Repository

- Developed an FP-Growth association rule mining based anomaly detection pipeline through streaming of sliding windows of size 6, stride 3 and overlap of 50% in order to keep the detection time delay below 30 seconds.
- Employed frequency-weighted score calculations with rule filtering criteria and validation of rule stability by using chronological N -fold cross validation.
- Compared the results with 3 unsupervised approaches (Isolation Forest, K-Means and DBSCAN) using feature sets by using agreement matrices and ROC graphs.

CERTIFICATIONS

- IIT KGP AI4ICPS – Advanced Machine Learning & Industrial Cyber-Physical Systems
- Smart Interviews – Data Structures, Algorithms & System Design